STAT230A Lecture 16 Leverage

1 Leverage

1.1 Definition: Leverage

令 hat matrix 为 $H = X(X^\top X)^{-1}X^\top \in \mathbb{R}^{n\times n}$, 其 elements 为 h_{ij}

则 subject i 的 **leverage score** 被定义为 h_{ii}, 即 H 的 $i\text{-th}$ diagonal element

1.2 Property: Leverage h_{ii} 的性质

性质 1:

由于 h_{ii} 为 H 的 $(i,i)\text{-th}$ element, 且 $(X^\top X)^{-1} \succeq 0$, 有

$$\boxed{h_{ii} = x_i^\top (X^\top X)^{-1} x_i \geq 0}$$

性质 2:

若 H 为 projection matrix of rank p (X 为 full-rank), 则

$$\begin{aligned} &\sum_{i=1}^n h_{ii} = \operatorname{Tr}(H) = \operatorname{rank}(H) = p \\ \Rightarrow \quad &\boxed{\frac{1}{n}\sum_{i=1}^n h_{ii} = \frac{p}{n}} \end{aligned}$$

性质 3:

由于 $H = H^2$, 有

$$\begin{aligned} h_{ii} =& \begin{bmatrix} h_{i1} & \cdots & h_{in} \end{bmatrix} \begin{bmatrix} h_{1i} \\ \vdots \\ h_{ni} \end{bmatrix} \\ =& \sum_{j=1}^n h_{ij} h_{ji} \end{aligned}$$

且由于 $H = H^\top$, 即 $h_{ij} = h_{ji}$, 因此

$$h_{ii} = \sum_{j=1}^n h_{ij}^2 = h_{ii}^2 + \sum_{j\neq i} h_{ij}^2$$

因此 $h_{ii} \geq h_{ii}^2$, 由于 $h_{ii} \geq 0$, 有

$$\boxed{h_{ii} \in [0,1]}$$

性质 4:

由于 $\hat{Y} = HY$, 有

$$\boxed{\hat{y}_i = \sum_{j=1}^n h_{ij} y_j = \underbrace{y_i h_{ii}}_{\text{self-weight}} + \sum_{j\neq i} h_{ij} y_j}$$

Logic

对于 Gauss-Markov Model, 有

$$\text{Cov}(\hat{Y}) = \sigma^2 H$$

因此

$$\text{Var}[\hat{y}_i] = \sigma^2 h_{ii}$$

这揭示了 leverage 名字的由来

1.3 Theorem: Leverage Decomposition

令:

- Regression 为 $Y \sim X$
- Design matrix 为 $X = (\vec{1} \quad X^*)$

则:

$$h_{ii} = \frac{1}{n} + \frac{D_i^2}{n-1} \geq \frac{1}{n}$$

其中:

- $D_i^2 = (x_i^* - \bar{x}^*)^\top S^{-1} (x_i^* - \bar{x}^*)$
- $S = \frac{1}{n-1} \sum_{i=1}^n (x_i^* - \bar{x}^*)(x_i^* - \bar{x}^*)^\top$ 为 X^* 的 sample covariance matrix
- $\bar{x}^* \in \mathbb{R}^{p-1}$ 为 X^* 的 all n rows 的 sample mean

Remark: 对 D_i 的理解

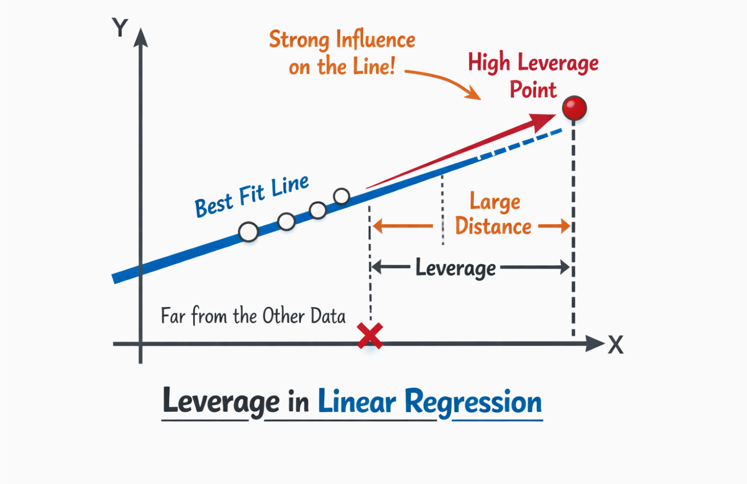
x_i^* 和 $\bar{x}^*$ 之间的 Euclidean distance 为

$$\sqrt{(x_i^* - \bar{x}^*)^\top (x_i^* - \bar{x}^*)}$$

则 D_i 可以理解为该 distance 的 normalized (more robust) version (Mahalanobis distance)

Remark: Leverage 的 Intuition

由此可知, 离其他 data points 越远的 data point, 其 leverage 越高, 这也是 leverage (物理学中意为 "杠杆") 名称的由来



Proof

对于 X^* 的 sample covariance matrix S, 其可被改写为:

$$
\begin{aligned}
S &= \frac{1}{n-1}\sum_{i=1}^{n}(x_i^* - \bar{x}^*)(x_i^* - \bar{x}^*)^\top \\
&= \frac{1}{n-1}(x^*)^\top (I - H_1) x^* \qquad \left(H_1 = \frac{1}{n}\vec{1}\vec{1}^\top \text{ 为 projection into } \vec{1}\right)
\end{aligned}
$$

因此对于 D_i^2, 其为 diagonal element of

$$
\begin{aligned}
&\begin{bmatrix}(x_1^*)^\top - (\bar{x}^*)^\top \\ \vdots \\ (x_n^*)^\top - (\bar{x}^*)^\top\end{bmatrix} S^{-1}\begin{bmatrix}x_1^* - \bar{x}^* & \cdots & x_n^* - \bar{x}^*\end{bmatrix} \\
=&(n-1)(I-H_1)x^*[(x^*)^\top(1-H_1)x^*]^{-1}(x^*)^\top(I-H_1) \\
=&(n-1)\tilde{x}^*[(\tilde{x}^*)^\top\tilde{x}^*]^{-1}(\tilde{x}^*)^\top \qquad (\tilde{x}^* := (I-H_1)x^*) \\
=&(n-1)\tilde{H} \qquad (\tilde{H} \text{ 为 projection into } \mathcal{C}(\tilde{X}^*)) \\
=&(n-1)(H-H_1)
\end{aligned}
$$

由此可以得出

$$
\begin{aligned}
& D_i^2 = (n-1)\left(h_{ii} - \frac{1}{n}\right) \\
\Rightarrow\quad & h_{ii} = \frac{1}{n} + \frac{D_i^2}{n-1}
\end{aligned}
$$

1.4 Theorem: Leave-One-Out Formula (LOO Formula)

令:

- $\hat{\beta}$ 为使用 full data 得到的 regression coefficient
- $\hat{\beta}_{[-i]}$ 为去除 observation i 后得到的 regression coefficient
- $\hat{\varepsilon}_i$ 为使用 full data 得到的 observation i 的 regression residual

则:

$$
\hat{\beta}_{[-i]} = \hat{\beta} - \frac{1}{1-h_{ii}}(X^\top X)^{-1}x_i\hat{\varepsilon}_i
$$

Remark: Leverage 和 Residual 对 Regression Coefficient 的影响

若 observation 对 regression coefficient 的影响很大, 则其可能满足:

- high leverage
- high residual

Proof

由于:

- $\hat{\beta} = (X^\top X)^{-1}X^\top Y$ 为 function of X 和 Y
- $X_{[-i]} = \begin{bmatrix} x_1^\top \\ \vdots \\ x_{i-1}^\top \\ x_{i+1}^\top \\ \vdots \\ x_n^\top \end{bmatrix} \in \mathbb{R}^{(n-1)\times p}$
- $Y_{[-i]} \in \begin{bmatrix} y_1 \\ \vdots \\ y_{i-1} \\ y_{i+1} \\ \vdots \\ y_n \end{bmatrix} \mathbb{R}^{n-1}$

则对于 $(X_{[-i]}^\top X_{[-i]})^{-1}$, 由于 $X^\top X = \sum_{i=1}^n x_i x_i^\top$, 使用 Sherman-Morrison Theorem, 其可被改写为:

$$
\begin{aligned}
(X_{[-i]}^\top X_{[-i]})^{-1} &= (X^\top X - x_i^\top x_i)^{-1} \\
&= (X^\top X)^{-1} + \frac{1}{1 - x_i^\top (X^\top X)^{-1} x_i} (X^\top X)^{-1} x_i x_i^\top (X^\top X)^{-1} \\
&= (X^\top X)^{-1} + \frac{1}{1 - h_{ii}} (X^\top X)^{-1} x_i x_i^\top (X^\top X)^{-1}
\end{aligned}
$$

因此 $\hat{\beta}_{[-i]} = (X_{[-i]}^\top X_{[-i]})^{-1} X_{[-i]}^\top Y_{[-i]}$ 可被改写为:

$$
\begin{aligned}
\hat{\beta}_{[-i]} \ &= \left((X^\top X)^{-1} + \frac{1}{1 - h_{ii}} (X^\top X)^{-1} x_i x_i^\top (X^\top X)^{-1} \right) [X^\top Y - x_i y_i] \\
&= (X^\top X)^{-1} X^\top Y \\
&\quad - (X^\top X)^{-1} x_i y_i \\
&\quad + \frac{1}{1 - h_{ii}} (X^\top X)^{-1} x_i \underbrace{x_i^\top (X^\top X)^{-1} X^\top Y}_{\hat{y}_i} \\
&\quad - \frac{1}{1 - h_{ii}} (X^\top X)^{-1} x_i \underbrace{x_i^\top (X^\top X)^{-1} x_i}_{h_{ii}} y_i \\
&= \hat{\beta} - (X^\top X)^{-1} x_i y_i + \frac{1}{1 - h_{ii}} (X^\top X)^{-1} x_i \hat{y}_i - \frac{h_{ii}}{1 - h_{ii}} (X^\top X)^{-1} x_i y_i \\
&= \hat{\beta} - \underbrace{\left(1 + \frac{h_{ii}}{1 - h_{ii}} \right)}_{\frac{1}{1 - h_{ii}}} (X^\top X)^{-1} x_i y_i + \left(\frac{1}{1 - h_{ii}} \right) (X^\top X)^{-1} x_i \hat{y}_i \\
&= \hat{\beta} - \frac{1}{1 - h_{ii}} (X^\top X)^{-1} x_i \hat{\varepsilon}_i
\end{aligned}
$$

⚠ Remark: Sherman-Morrison Theorem

Sherman-Morrison Theorem 可用于计算 special case of block matrix inversion:

$$
(A + uv^\top)^{-1} = A^{-1} - \frac{1}{1 + v^\top A^{-1} u} A^{-1} u v^\top A^{-1}
$$